Exam Computer Networks

May 27, 2025

Name: …………………………………………… Student number: ………………......……

General remarks

- Fill out your name and student number at the top of each page.
- Place your student card visibly on your desk.
- You may bring 1 double-sided A4 sheet with notes.
- No other course material, notes, or book may be consulted during the exam.
- The use of a cell phone or any other means of communication is strictly forbidden.
- Always motivate your answer. Show how you came to your results.
- The expected size of the answer is in line with the size of the provided answer box.
- If your answer does not fit within the box, ask for an extra piece of paper. Clearly mark that your answer continues elsewhere.
- Plan your time well, you have **3 hours**.
- You can use a calculator.
- You are free to provide answers either in English or in Dutch.

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Question 1: HTTP Performance

1. What is the difference between persistent and non-persistent HTTP connections? Show the difference using a sequence diagram. What is the difference between the two in terms of performance?

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2. Consider an HTTP/1.1 client that visits a website. The round-trip time (RTT) between the client and web server equals 150 ms. The client requests an HTML webpage that contains 5 images. The transmission time of the HTML webpage is 50 ms, while that of each image is 200 ms. The transmission time of HTTP requests and TCP control packets is considered negligible. You may assume the client knows the IP address of the web server. How much time does it take in total to request the webpage and images:
   a. Non-persistent HTTP with no parallel TCP connections?
   b. Non-persistent HTTP with 4 parallel TCP connections?
   c. Persistent HTTP with no parallel TCP connections?

3. What additional steps are needed, at the transport layer and above, if the HTTP client only knows the hostname and not the IP address of the web server?

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Question 2: Transport Control Protocol (TCP)

TCP keeps track of an estimate round trip time (RTT) between the sender and receiver. Each time an RTT measurement is collected by the sender (i.e., SampleRTT), the estimated RTT (i.e., EstimatedRTT) and its deviation (i.e., DevRTT) are calculated as follows:

$$\text{EstimatedRTT} = (1-\alpha)\cdot\text{EstimatedRTT} + \alpha\cdot\text{SampleRTT}$$
$$\text{DevRTT} = (1-\beta)\cdot\text{DevRTT} + \beta\cdot|\text{SampleRTT} - \text{EstimatedRTT}|$$

1. When the TCP Timestap option is not used, TCP will not update the EstimatedRTT and DevRTT values for segments that are retransmissions. Why is that?

2. What are the EstimatedRTT and DevRTT values used for by TCP? What happens if the estimated RTT differs from the real RTT (i.e., it is higher or lower)?

3. What is fast retransmission? Why is it used, and what are its advantages?

4. What is the goal of flow control? Show how it works in TCP using a diagram.

Question 3: CSMA/CD

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) is random channel access protocol. Its efficiency (assuming a large number of transmitters) is defined as a function of both the propagation delay (d_{prop}) and maximum frame transmission time (d_{trans}):

$$\text{Efficiency} = \frac{1}{1 + 5\frac{d_{prop}}{d_{trans}}}$$

1. How is the efficiency of CSMA/CD affected by changes in the link length and link data rate? Explain why.

2. You may assume a maximum frame size of 1500 bytes, and a propagation speed of 200 000 km/s. Calculate the efficiency of CSMA/CD for a link length of 10 m, and data rates 10 Gbps and 40 Gbps. How can efficiency be improved without changing the data rate?

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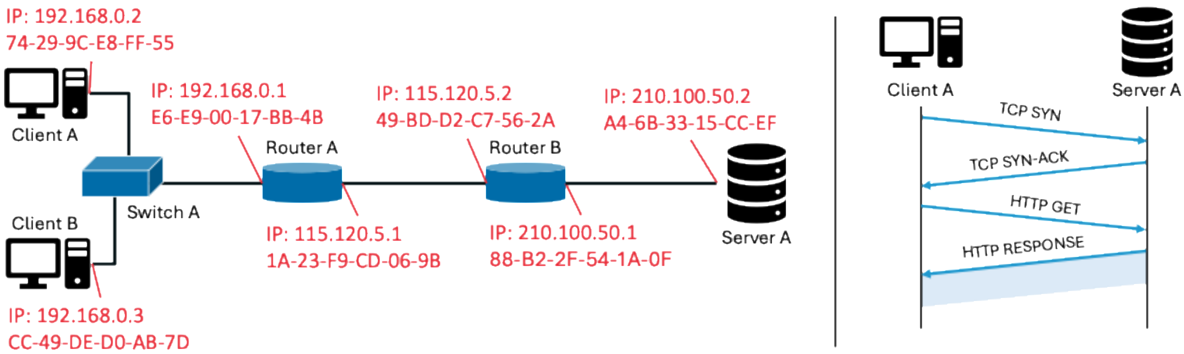
Question 4: End-to-end HTTP transmission

Consider the scenario shown in the figure below. The left part of the figure shows the network topology, consisting of 2 clients, a switch, 2 routers, and a server. Client A initiates a file download from Server A via HTTP, as shown in the right part of the figure. You may assume that IP addresses in the range 192.168.0.0/24 are private addresses, while all others are public. Moreover, all subnetworks have a 24-bit subnet mask.



1. Assume the HTTP message exchange shown on the right has already taken place. Client A uses TCP port 5421 and Server A uses TCP port 443. What are the contents of the Network Address Translation (NAT) table of Routers A and B?

2. What is the source and destination IP and MAC address of the HTTP GET request sent by Client A, when it is (1) sent by Client A, and (2) received by Server A?

3. Assume that Client A does not know the IP and MAC address of Router A at the time it wants to send the TCP SYN packet to Server A. What steps does Client A need to take to discover both addresses?

4. Assume Switch A has an empty Switch table before the TCP SYN packet is sent by Client A to Server A. How does Switch A forward the TCP SYN packet? What is the content of the Switch table after the client receives the HTTP RESPONSE?